

DEFEATING POLLUTION WITH THE IoT: Solutions In India

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Indian cities have been struggling to bring pollution levels in check. Recent World Health Organisation reports have revealed that India has the most polluted air in the world—cities like Kanpur, Mumbai, New Delhi and Varanasi, among others, appear on the top of the list. New Delhi has often been in the red alert radar, like last year, when the city's air quality crossed 70 times the safety limit.

Apart from air, there is concern regarding water pollution. India's major rivers—Ganga, Yamuna and Narmada—and most other water bodies are heavily polluted.

While air pollution is mostly contributed by dust from unregulated construction works, emission from vehicles, industries and homes, and so on, water pollution is caused due to unregulated waste dumped into water bodies that may come from

households or factories, poor sewage management, non-invigilated washing and bathing by people, etc. We must include soil pollution as well to get a clearer picture of where we stand.

To bring these critical environmental issues in control, experts are arming themselves with modern technologies like the Internet of Things (IoT). The IoT takes the role of an enabler, providing clear visibility of real-time environmental conditions to users and allowing them to make necessary changes to improve conditions. Let us look at how the IoT is helping people mitigate environmental problems.

How the IoT reduces pollution

Smart devices powered by the IoT gives users the opportunity to optimise the very aspects

of their operations that may improve the impact on the environment. Ashish Mathur, managing director, Jamshedpur Utility and Services Co. (JUSCO), says, "IoT setups provide a clear dashboard-based compact view of all necessary data to decision makers. With information like PM levels in the air, moisture, humidity, pH of water and so on being available in one place, making strategies to mitigate air pollution becomes easy."

The IoT is embedded in every smart appliance today and, consequently, helps mitigate environmental issues in one way or the other. Amarjeet Singh, chief technology officer, Zenatix Solutions Pvt Ltd, says, "IoT solutions can handle environmental issues directly or indirectly. Solutions like air quality or environment monitoring systems take a direct approach by displaying live conditions of the environment, be it indoors or outdoors.

"As a result, users know what can be done to control the readings, if these are above safety levels. Other solutions bring indirect benefits to the environment, such as technology that reduces energy consumption. If homes, industries and cities can optimise their energy-driven appliances to reduce electricity usage, a massive reduction in carbon footprint, thermal emission and, subsequently, pollution can be achieved."

Industries are one of the largest contributors to environmental pollution. Hence, many big factories are deploying IoT solutions in their premises. Amiya Kumar Samantaray, chief executive officer and founder, Phoenix Robotix Pvt Ltd, says, "Central Pollution Control Board (CPCB) has come up with guidelines for industries and cities to monitor pollution levels online. All data gets collected on a centralised platform, where it is analysed and risk areas are highlighted. With remote access to all analytics, industries are better able to prepare for pollution and, eventually, follow CPCB norms better."

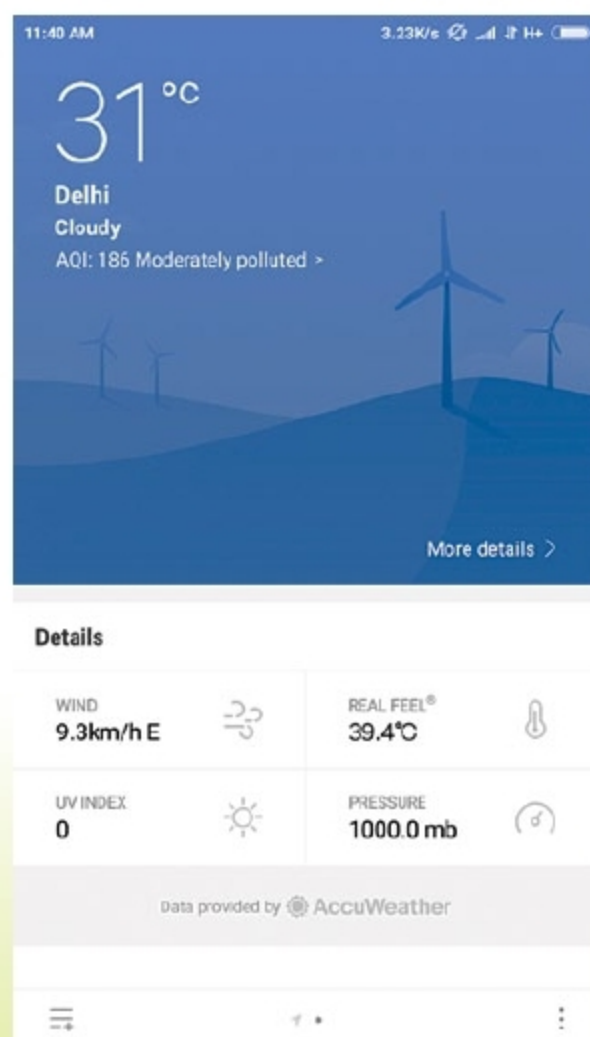


Fig. 1: Smartphones display the air quality index of regions